

CAMPUS

The Master Plan

Vision, game design, everything left to do,
architecture for millions, pinch points,
milestones and the grand prompt for any AI.

REJUVINI CAMPUS · Prepared for John

23 October 2026 · Version 1.0 · Estimates are labelled; evidence beats opinion

Contents

The Vision and Where We Stand 4

Your explicit asks, and where each one stands 4

Completion by area (last audit, 12 Oct, plus what changed since) 4

Game Design Bible 6

What the best mobile games get right, and the CAMPUS answer 6

Six pillars (every feature must serve at least one) 6

Player journey 7

Core loops (nested, every loop feeds the next) 7

Mode and content catalogue 8

Progression and economy 9

Campus Wars design (the competitive spine) 9

Live-ops rhythm 10

Feel and polish requirements (treat as scope, not garnish) 10

Everything That Has To Be Done 11

A. Foundation and release health 11

B. App shell, onboarding, HUD, controls 11

C. Renderer and graphics 12

D. Asset pipeline and content 13

E. World: zones, buildings, interiors 14

F. Avatars, NPCs, animation 15

G. Gameplay and activities 15

H. Realtime multiplayer 16

I. Social, events, clubs 16

J. Campus Wars and progression 17

K. Economy, shop, housing, payments 17

L. Creator Studio and AI helpers 17

M. Safety, moderation, legal 18

N. Backend and data 18

O. Performance, analytics, observability 19

P. Security and anti-cheat 19

Q. QA and evidence 20

R. DevOps, release, cost . . . 20
S. Audio, localisation, accessibility . . . 21
T. Growth, community, live ops, support . . . 21

Architecture for Millions . . . 22
4.1 Planning targets and capacity . . . 22
4.2 The whole system . . . 23
4.3 Client architecture . . . 24
4.4 Realtime architecture . . . 25
4.5 Reward authority and anti-cheat . . . 25
4.6 Data architecture . . . 26
4.7 Security and privacy architecture . . . 26
4.8 Infrastructure and regions . . . 27
4.9 Scaling ladder . . . 27
4.10 Service level objectives . . . 28
4.11 Performance budgets . . . 28
4.12 Target repository structure . . . 28

Pinch Points . . . 30

Milestones . . . 32
Release ladder . . . 32
Go / no-go gates (from your docs) mapped to milestones . . . 32
The milestones in detail . . . 33

Your Next Ten Days . . . 38
Decisions only you can make . . . 38

The Grand Prompt . . . 39

Quick Reference . . . 42
Technology choices at a glance . . . 42
What not to build before V1 . . . 42
The thirty-second fun test . . . 43

Closing: The Bar . . . 43

PART 1

The Vision and Where We Stand

CAMPUS is a browser-first Nigerian campus life game. You walk through the Grand Gate of **Rejuvini Campus**, build an avatar and live the campus: short activities, a real 3D library, jobs, clubs, events, a room to decorate, and points for your campus in **Campus Wars**. It has to run on a $100 Android phone, over mobile data, in a browser tab, and still look like a premium game.

> **The one sentence to protect:** a student opens a link, is in the gate scene within seconds, is having fun in 30 seconds, and has a reason to come back tomorrow and fight for their campus every week.

Your explicit asks, and where each one stands

Your ask (your words, condensed)	Status today	Honest note
Browser-first Nigerian campus sim, built from the Master Implementation Prompt	Built ~68%, verified ~36%	Numbers from the 12-13 Oct audit (±5 pts). Nothing is formally PASS yet.
Free Fire / CoD class graphics, sharp on phones	Partly. Gap is large	Custom WebGL renderer works but is heuristic, untyped and uncompressed. See Part 5 (renderer) and M3-M4.
Real enterable Library: door, books in every field, stairs, downstairs, Pick, sit and read	Built (LibraryWorld), unverified	Needs real-phone proof, content depth and server-side reading rewards.
Single grand gate named Rejuvini Campus, your logo, smooth realistic police and pass	Built, mostly seen in screenshots	Card grip, thumbs-up frame and a real-phone check are still open (handoff note).
Campus Wars seasons, events, creator studio, fictional Campus Coins, no gambling	Wars + events built. Creator 0%	Creator Studio and AI helpers are deliberately after V1.
Deploy on Render, hand back small zips, notes for another AI	Done (staging, free tier)	Free tier sleeps and has no persistent disk: fine for testing, not for players.
Be able to serve millions	Not yet	Needs Postgres, Redis, signed grants, a renderer upgrade and a load-tested realtime tier. All specified in Part 4.

Completion by area (last audit, 12 Oct, plus what changed since)

Area	Wt	Built	Verified	Biggest gap
Realtime multiplayer	7	92%	78%	Binary protocol, sharding, 10k CCU load test
Campus Wars and progression	5	75%	45%	Fairness maths, season ceremony, appeals
Activities and mini-games	6	65%	45%	Server-authority review of shootout/delivery/rhythm
3D renderer, camera, lighting	7	75%	30%	Typed WebGL2 core, KTX2, LOD, real-phone FPS
All 22 zones	8	65%	20%	Hero quality for every zone, interiors, day/night
Avatars, NPCs, animation	5	55%	15%	GPU skinning, crowds, faces/hands, sync proof
Safety, moderation, legal	6	50%	25%	Moderator console, appeals, NDPA, age policy
Backend, data, ops	5	70%	40%	Postgres adapter, Redis, backups rehearsed
Performance, analytics, infra	6	50%	20%	Real-device table, funnels, cost controls
Creator Studio and AI	4	0%	0%	Everything (after V1)
Audio, localisation, accessibility	n/a	~0%	0%	Not in the old checklist weights. Added in this plan

What I could and could not read. I read your saved notes, every doc in the repo (status audit, road-to-finished, vision gaps, graphics vision, handoffs, architecture, decisions) and the code layout. The Master Implementation Prompt, the Phase 4+ Addendum, the Art Direction prompt and the 1,081-item Master Build Checklist are **not in the repo**. I reconciled to their summaries in the docs. Put them in `docs/vision/` and the plan below can be reconciled line by line.

PART 2: ARRANGED LIKE A LIVE GAME, NOT A DEMO

Game Design Bible

You asked me to arrange CAMPUS as if designing a game better than Free Fire. Free Fire is not better because of its map. It wins on **session design, device reach, rhythm, identity and live operations**. This part takes each lesson and answers it in campus terms. CAMPUS is a social sim, so the answer is not a battle royale; it is the same machine of habit, status and mastery, pointed at campus life.

What the best mobile games get right, and the CAMPUS answer

Lesson	How winners do it	CAMPUS answer
Plays on weak phones	Aggressive optimisation, tiers, small downloads, regional tuning	Three quality tiers with auto downshift, 3 MB first load, Save-Data mode, 30 FPS floor, tested on Tecno/Infinix/Samsung A-class devices
Fast sessions	~10 min complete loop, instant matchmaking	First fun in 30 s; a satisfying session is 8-12 min; every activity is 1-4 min
Identity and status	Characters, cosmetics, ranks, titles	Campus pride (colours, crest, chants), avatar wardrobe, titles, a room to show off, season badges
Squad play	Friends are the retention engine	Clubs (groups), 2-4 player squads for team activities, friend-follow into instances, club leaderboards
Ranked seasons	Fresh ladder every few weeks, rewards on a track	12-week Campus Wars seasons with leagues, season pass track, ceremony and a clean reset
Event cadence	Always something new this week	Weekly rhythm calendar, monthly themes, Nigerian academic-calendar events
Creators and spectating	Streamers, replays, community content	Spectate matches, shareable highlight cards, Creator Studio after V1
Trust and fairness	Anti-cheat, clear rules, fast bans	Server authority, per-capita scoring, anomaly detection, appeals
Feel (juice)	Every tap answers with motion, sound, haptics	Audio and haptics layer, hit-stop, particles, camera polish. Treated as a workstream, not decoration

Six pillars (every feature must serve at least one)

Pillar	Promise to the player	Proof we hold it
1 Belong	My campus is my team, my place, my pride	Campus identity in gate, HUD, avatars, wars
2 Play	There is always a quick, fun thing to do	Activity starts within 3 taps; no dead time
3 Compete	My effort moves my campus up a real ladder	Fair, explainable, resettable Campus Wars
4 Express	I can look and live the way I want	Wardrobe, room editor, emotes, titles, later creator tools
5 Learn and earn	Time here makes me better off	Study rewards in the library, jobs, quiz bowls by subject, real skills
6 Safe and fair	Nobody cheats me, nobody harasses me	Moderation, reporting, appeals, age-aware defaults

Player journey

Moment	Player goal	What the game must do	Success metric
0-30 s	See it, feel it	Gate vista, soft music, big Rejuvini Campus sign, avatar already moving	Load to first input under 8 s on throttled 4G
30-90 s	Be let in	Calm guards walk up facing you, show pass, wicket opens, welcome	Entry completes without confusion; skip is always available
1-5 min	First win	Guided first activity, first Coins, first Points for my campus	Reward granted once, ledger exact, visible on HUD
5-12 min	Reason to stay	Meet someone, see my campus rank, unlock one cosmetic	Friend request or club join in first session
Day 1 return	Continue	Daily mission, event reminder, what changed since you left	D1 retention target 35-40%
Day 7	Habit	Club, weekly events, wars standings, room upgrades	D7 target 15-20%
Day 30	Identity	Season track, titles, creator/club status	D30 target 8-10%

Targets are planning goals for casual mobile social games, not measurements. We replace them with real funnel data from M13.

Core loops (nested, every loop feeds the next)

Loop	Length	Action	Reward	Feeds
Micro	10–30 s	Move, interact, emote, pick an item	Juice: sound, motion, small XP	Flow
Activity	1–4 min	Quiz, reaction, hunt, shootout, delivery, rhythm, shift	Coins + Campus Points + XP	Session
Session	8–15 min	3–5 activities, one social moment, one goal progress	Daily mission progress, streak	Day
Daily	1 day	Mission set, streak, friend nudge	Season XP, Coins, cosmetic shards (not random)	Week
Weekly	7 days	Event night, club challenge, wars round	Season track tiers, titles, trophies	Season
Season	12 weeks	Campus Wars league climb	Season pass items, crest, ceremony, hall of fame	Identity

Mode and content catalogue

Mode	What it is	Today	Target
Free roam campus	Walk 22 zones, enter buildings, meet players and NPCs	Built	Hero polish + interiors + audio
Quiz	Timed subject quizzes, solo and bowl format	API + UI built	Subject banks, quiz bowl rooms (5v5)
Reaction	Server-timed reaction test	Built	Duels and leaderboards
Scavenger hunt	Checkpoint trail with clues	Built	Creator templates later
Penalty shootout	Football duel, server-judged	Logic built, review pending	Live PvP match rooms at 30 Hz
Delivery runs	Timed campus courier job	Logic built	Squad relay mode
Rhythm	Beat-matching mini-game	Logic built	Music packs, club performances
Jobs and shifts	Part-time jobs with shift tasks	Built	More jobs, career tracks
Library study	Enter, pick books, sit and read	Built (v18), unverified	Real content, reading goals, study rooms with friends
Events	Scheduled campus events with RSVP and check-in	Built	Live in-world event stages
Clubs and squads	Groups, join, kick, leave	Built	Club halls, club wars, ranks
Home and room	Room editor, shop items	Built	Visitors, showcase, contests
Campus Wars	Seasonal campus ladder	Built	Leagues, ceremony, appeals, brackets
Inter-Campus Cup	Knockout tournament across campuses	Missing	Quarterly flagship event, spectate mode

Mode	What it is	Today	Target
Creator Studio	Safe templates for quests, events, rooms	Missing	After V1, moderated and rate-limited

Progression and economy

Three currencies only, each with one job, so the economy stays explainable and auditable:

Currency	Earn	Spend	Rules
Campus Coins (soft)	Activities, jobs, events, daily missions	Shop cosmetics, room items, club upgrades	Daily earn cap with diminishing returns; every grant is an idempotent ledger entry
Campus Points	Activities and events, tagged to your campus	Not spendable	Feeds Campus Wars only; per-account daily contribution cap
Rejuvini Credits (premium)	Bought with NGN via Paystack/Flutterwave, plus rare season pass drops	Cosmetics and season pass only	Direct purchase of a visible item, never random; refundable window; separate ledger account

- Sources must be capped and sinks must be attractive. Review the faucet/sink ratio every season from ledger data.
- No item is better at winning. Cosmetics, titles, rooms and convenience only.
- Every purchase shows the exact item and price in Naira before the pay button.
- Season pass: free track plus premium track. Both are fair; premium is cosmetic and convenience.

Campus Wars design (the competitive spine)

- **Season:** 12 weeks, 3-week phases with a themed week each phase, a final week with the Inter-Campus Cup.
- **Fairness:** standings use per-capita active-player score, not raw totals, with a contributor floor so a campus needs enough participants to rank. Per-account daily cap stops grinders and multi-accounting from dominating.
- **Leagues:** campuses are grouped by active size (Bronze to Champion). Promotion and relegation at season end so small campuses always have a winnable ladder.
- **Transparency:** a Wars screen shows how points are computed, last-hour changes and your personal contribution.
- **Integrity:** anomaly detection on contribution spikes, device and behaviour clustering, freeze-and-review before any rewards are paid, appeal path with a human reviewer.
- **Ceremony:** a 3D awards stage in the gate forecourt, season recap card (shareable image), permanent hall of fame.

Live-ops rhythm

When	What	Why
Daily	3 missions, a rotating featured activity, streak	Reason to open the game
Mon	Wars round opens, club challenge drops	Weekly goal
Wed	Quiz bowl night	Learning plus social
Fri	Social night: music, emote party, shop refresh	Peak hours
Sat	Sports day: shootout and relay brackets	Competition
Sun	Wars recap, rewards, new mission set	Closure and anticipation
Monthly	Themed event: matriculation, freshers week, inter-house, culture day	Freshness
Season	New theme, new pass, wars ceremony	Reset and prestige

> Tie the calendar to the Nigerian academic year: resumption and freshers, matriculation, exams (study rewards in the library), inter-house sports, convocation, festive breaks. That is a hook no foreign game can copy.

Feel and polish requirements (treat as scope, not garnish)

- Audio: ambient bed per zone, footsteps by surface, UI sounds, music with a ducking mixer, positional voices. Sprite sheets under 1 MB for first load.
- Haptics on key actions (Vibration API) with an off switch.
- Camera: smooth chase, no wall clipping, comfortable portrait framing, subtle sprint FOV.
- UI motion under 150 ms, consistent tokens, thumb-zone controls: left stick, right action cluster, drag-to-look.
- Everything has loading, empty, error and offline states. No dead buttons.
- Nigerian voice: Pidgin NPC lines, local food, music, campus slang, dance emotes (originals, no copied choreography).

PART 3: THE COMPLETE INVENTORY

Everything That Has To Be Done

Twenty workstreams, grouped so one person or one AI can own each. Every line has a size and a milestone. **Size** = focused effort for one strong developer working with AI help: **S** up to 2 days, **M** up to a week, **L** up to 2 weeks, **XL** more than 2 weeks. Milestones are defined in Part 6.

> **175 tasks in 20 workstreams.** That is the honest size of finishing CAMPUS properly. It is large, which is why Part 6 orders it so the game is playable, safe and testable at every stage instead of all-or-nothing.

A. Foundation and release health

Goal: Repo builds clean from scratch, CI is green, nothing is claimed without proof.

ID	Task	Size	When
A01	Regenerate and commit pnpm-lock.yaml (stale: three, vitest); restore frozen-lockfile	S	M0
A02	Run typecheck, lint, API tests (app, social, jobs-wars), next build; fix all failures	M	M0
A03	CI pipeline: install, typecheck, lint, unit, build, asset audit, 3D checks on every push	M	M0
A04	Repair known red checks: check:3d missing tests, hair_ponytail, tri-budget alley/boulevard, check:vista	M	M0
A05	Move vision docs (Master Prompt, Addendum, Art Direction, Checklist) into docs/vision/	S	M0
A06	Single status board: checklist items with PASS/PARTIAL/FAIL and evidence links	M	M0
A07	ADR habit: every non-trivial decision in DECISIONS.md; handoff note updated per session	S	M0
A08	Branch policy: main always green, short-lived feature branches, tagged releases	S	M0
A09	Dependency hygiene: Renovate/Dependabot, license scan, SBOM	M	M14

B. App shell, onboarding, HUD, controls

Goal: A first-timer is playing in under a minute and never feels lost. Portrait-first.

ID	Task	Size	When
B01	Landing to /start to avatar builder to gate flow without dead ends; resume if refreshed	M	M1
B02	Guest-first sign-in with account upgrade later (no wall before fun)	M	M1
B03	Gate UI/Pass card carry Rejuvini Campus name and logo; skip intro always visible	S	M1
B04	Portrait controls: left stick, right action cluster, drag camera, sprint, interact, emote wheel	L	M1
B05	HUD from shared tokens: coins, points, mission, minimap, chat, notifications, phone menu	L	M1
B06	First-session guided path: first activity, first reward, first social prompt	M	M1
B07	What-changed-since-you-left screen and daily missions	M	M9
B08	Settings: quality, Save-Data, sound, haptics, language, controls, reduced motion	M	M12
B09	Responsive captures 360x800, 390x844, 720x1640, 1440x900 with PASS/PARTIAL/FAIL	M	M1
B10	PWA: install prompt, offline shell, update flow; later Play Store wrapper (TWA/Capacitor)	L	M14

C. Renderer and graphics

Goal: Free Fire-class stylised-realism at 30 FPS on low Android; typed, tested, maintainable.

ID	Task	Size	When
C01	Time-boxed spike: upgrade custom renderer vs adopt Three.js on the same Gate scene, on a real phone	L	M3
C02	Adopt WebGL2 core with per-draw material IDs (kill colour/normal heuristics in shaders)	XL	M3
C03	Remove @ts-nocheck: strict types for renderer, shaders, loaders	L	M3
C04	glTF pipeline: meshopt geometry, KTX2/Basis textures, mipmaps, atlas shared materials	L	M3
C05	Instancing and batching for props/foliage; GPU skinning; crowd instancing for far NPCs	XL	M5
C06	Chunked zone streaming, frustum + grid culling, HLOD and LOD fade (no popping)	XL	M4
C07	Lighting: baked AO and lightmaps for hero zones, tight single-cascade sun shadows, hemisphere fill	L	M4
C08	Sky, fog, day/night from Lagos time; optional bloom for signage on High tier only	M	M4
C09	Quality tiers + auto downshift on FPS/thermal; 30 FPS cap mode; resolution scaling	L	M3
C10	WebGL context-loss recovery and 2D fallback proven by test	M	M3
C11	Hero shaders: wet/dust paving, foliage wind, glass, emissive signage with consistent PBR-lite	L	M4
C12	WebGPU as progressive enhancement only after WebGL2 path is perfect	L	M16

D. Asset pipeline and content

Goal: Every asset is licensed, budgeted, compressed, registered and has a fallback.

ID	Task	Size	When
D01	Asset manifest as the single registry with source, licence, triangles, bytes, LOD, fallback	M	M3
D02	Automated import: unpack, normalise scale/pivot, LODs, compress, register, audit in CI	L	M3
D03	Filter Kenney library to campus-appropriate pieces; delete the rest from runtime	M	M4
D04	Hand-author hero kit: gate, student centre, library, lecture halls, hostels, cafeteria, sports ground	XL	M4
D05	Modular architecture kit: walls, windows, columns, canopies, stairs, railings, AC units, signage	XL	M4
D06	Real PBR sets (CC0) for paving, brick, roof, grass compressed to KTX2 512/1K	M	M4
D07	Vegetation set: palms, neem, flame trees, hedges, planters (real leaf cards)	L	M4
D08	Props pass for all 22 zones, human-scale check, no floating or clipping	XL	M12
D09	Texture/byte budgets enforced in CI; fail the build on overrun	M	M3
D10	Credits and licence file kept current (docs/CREDITS.md, ASSET_SOURCES.md)	S	M0

E. World: zones, buildings, interiors

Goal: 22 zones that each look intentional, are walkable, enterable and readable on a phone.

ID	Task	Size	When
E01	Grand Gate: finish card grip, thumbs-up, wicket swing check, police behaviour on real device	M	M4
E02	Boulevard and Student Commons hero polish; seating islands, cafe, fountain, lamps	XL	M4
E03	Library: real interiors, stair to lower level, books by field, Pick, sit-and-read, study rooms	XL	M4
E04	Walk all 158 enterable rooms: doors, interiors, collision, lighting	XL	M12
E05	Capture each zone close/medium/far at Low/Balanced/High; fix what is wrong	XL	M12
E06	Door system: human-like open/close animation, enter buttons, locked/open hours	L	M12
E07	Zone identity: unique silhouette landmark per major district	XL	M12
E08	Day/night and event dressing (matriculation banners, convocation stage, festive lights)	L	M12
E09	Server colliders match visible solids; props never walkable-through	L	M12
E10	Minimap and wayfinding: signage, route hints, auto-walk to a place	M	M12

F. Avatars, NPCs, animation

Goal: Characters that look good up close, move like humans and hold objects naturally.

ID	Task	Size	When
F01	GPU skinning plus animation texture crowds for far NPCs; CPU skinning only for hero close-ups	XL	M5
F02	Face, hair cards, hands with finger poses (grip card, point, wave, thumbs-up, hold book)	XL	M5
F03	Foot IK and speed-matched locomotion; stop foot sliding; seat height on real benches	L	M5
F04	Wardrobe system: outfits, shoes, accessories, skin tones, hair as data; fast rebuild	L	M5
F05	Emotes, dances, interaction animations (open door, sit, read, carry, throw ball)	L	M5
F06	NPC routines: collision-aware paths, schedules by time of day, reactions to the player	XL	M5
F07	NPC dialogue: deterministic scripted lines first, AI only later with caps and filters	M	M5
F08	Two-browser proof: animation synced from server-validated movement	M	M5
F09	Guard/police behaviour kit: face player, approach, greet, hand-to-hand, never stare sideways	M	M5

G. Gameplay and activities

Goal: A handful of loops so polished they are addictive; all outcomes server-judged.

ID	Task	Size	When
G01	Server-authority review of shootout, delivery, rhythm; move judging server-side	L	M6
G02	Quiz: subject banks (campus life, general, exam-prep subjects), difficulty tiers, anti-repeat	L	M6
G03	Reaction: duels and leaderboards with server timing and anti-bot	M	M6
G04	Scavenger hunt: more trails, hints, team mode	M	M6
G05	Penalty shootout as live PvP match room at 30 Hz	XL	M7
G06	Delivery and relay: squad mode	L	M6
G07	Rhythm: music packs, difficulty, offset calibration per device	L	M6
G08	Jobs: more roles, career track, shift variety, caps	M	M6
G09	Library study: reading goals, quiz-after-reading rewards, study rooms with friends	L	M6
G10	Juice pass on every activity: sound, haptics, particles, result screens, replay buttons	L	M6
G11	One-polished-loop rule: pick the top 3 activities and finish them before adding more	S	M6

H. Realtime multiplayer

Goal: 60-80 players per instance feel smooth on poor networks; thousands of instances run.

ID	Task	Size	When
H01	Binary versioned protocol (quantised positions, bit flags); keep JSON for debug	L	M7
H02	Interest management by spatial grid; delta snapshots; priority by distance	L	M7
H03	Client prediction and reconciliation tuned on 150-300 ms jitter; interpolation buffer 100-150 ms	L	M7
H04	Room allocator: instance per campus, friend-follow, overflow, party join	XL	M7
H05	Match rooms (sports, quiz bowl) spawned on demand with own tick	XL	M7
H06	Gateway tier (stateless WebSocket edge) separated from room workers	XL	M16
H07	Graceful drain, reconnect storms prevented with jittered backoff and admission control	L	M7
H08	Load and chaos tests: 1k, 10k, 100k simulated clients; tick overrun and RTT dashboards	XL	M7
H09	Server tick budget and load shedding (drop cosmetic updates before gameplay)	M	M7
H10	Cross-instance chat and presence over Redis pub/sub	L	M8
H11	Region placement: nearest room by measured RTT	L	M16

I. Social, events, clubs

Goal: Friends are the retention engine; every social feature is safe by default.

ID	Task	Size	When
I01	Friends, requests, blocks, mutes: server-side authority and two-account tests	M	M8
I02	Clubs: halls, ranks, club chat (moderated), club challenges	XL	M8
I03	Squads (2-4) for team activities with ready-check	L	M8
I04	Chat: quick-chat and emoji by default; text chat with filter, rate limits, report button	L	M8
I05	Events: RSVP/cancel, reminders, live stage in the world, post-event recap	L	M8
I06	Notifications: in-app first; web push opt-in; no spam, frequency caps	L	M8
I07	Invite links and referral with abuse limits	M	M8
I08	Presence and Find-friend with privacy controls	M	M8
I09	Spectate mode for matches and events	L	M16

J. Campus Wars and progression

Goal: A ladder everyone believes is fair, with a ceremony worth showing off.

ID	Task	Size	When
J01	Per-capita scoring, contributor floor, league brackets by size, promotion/relegation	L	M9
J02	Season lifecycle: create, run, freeze, review, reward, archive, reset (tested)	L	M9
J03	Leaderboards as Redis sorted sets with periodic Postgres snapshots	M	M9
J04	Anomaly detection and freeze-before-pay integrity checks	L	M9
J05	Season pass: free + premium tracks, XP sources, catch-up mechanic	L	M11
J06	Season ceremony stage, recap card, hall of fame	L	M9
J07	Explainable scoring screen and appeal path	M	M9
J08	Titles, badges, trophies with permanent history	M	M9
J09	Inter-Campus Cup tournament engine (bracket, rooms, spectate)	XL	M16

K. Economy, shop, housing, payments

Goal: Money-like state is exact; players trust prices and rewards.

ID	Task	Size	When
K01	Ledger hardening: double-entry accounts, monthly partitions, reconcile job, audit view	L	M2
K02	Faucet/sink dashboard and balance tuning from real data	M	M13
K03	Shop: catalogue service, featured rotation, NGN price display, item preview in 3D	L	M11
K04	Payments via Paystack/Flutterwave: checkout, webhook signature check, idempotent fulfilment, refunds	XL	M11
K05	Premium currency account separated from soft currency; fraud and chargeback handling	L	M11
K06	Room editor: save/load, visitors, showcase, item limits, moderation of names	L	M11
K07	Telecom/airtime billing partners for non-card users (research and pilot)	L	M14
K08	Sponsorship placements: labelled, capped, reviewed (after traffic is measurable)	L	M16
K09	Creator revenue share ledger and payout rules	L	M15

L. Creator Studio and AI helpers

Goal: Safe, declarative creation; AI as a bounded helper, never a risk.

ID	Task	Size	When
L01	Package format: declarative JSON validated by schema; server never runs user scripts	L	M15
L02	Templates: quiz, scavenger hunt, event, room decoration from approved catalogue	XL	M15
L03	Review pipeline: automatic checks, human moderation, quarantine, rollback	XL	M15
L04	Creator roles, quotas, rate limits, attribution and analytics (aggregate only)	L	M15
L05	AI NPC dialogue: short, cached, rate-limited, age-safe, deterministic fallback	L	M15
L06	AI content generation outputs structured data only, with spend ceilings	L	M15

M. Safety, moderation, legal

Goal: Safe enough to open to students, many of them minors.

ID	Task	Size	When
M01	Moderator console UI: queue, evidence, actions with reason, actor, timestamp, audit log	XL	M10
M02	Appeals with separate reviewer permissions; repeat-offender policy documented	L	M10
M03	Report flow everywhere; evidence kept minimal and access-controlled	M	M10
M04	Chat/name filters (Nigerian English and Pidgin lists), image/UGC review before publish	L	M10
M05	Age-aware defaults: strangers cannot DM, limited chat for under-18, parental info page	L	M10
M06	Privacy policy, terms, cookie notice written for Nigerian law; Nigerian counsel review	L	M10
M07	NDPA 2023: lawful basis, data inventory, DPIA, retention, deletion and export, NDPC registration check	L	M10
M08	Backup/restore and incident runbooks rehearsed	M	M10
M09	Admin MFA, least privilege, audit log retention	M	M10

N. Backend and data

Goal: No single machine holds anything important; data survives deploys and mistakes.

ID	Task	Size	When
N01	Postgres adapter behind the Store interface; migrations controlled and non-destructive	XL	M2
N02	Data migration from JSON snapshot; restore test	M	M2
N03	Redis: presence, rate limits, leaderboards, pub/sub, short-TTL keys	L	M2
N04	Signed proximity grants (HMAC/JWT) so API instances are stateless	M	M2
N05	Transactional outbox to queue (BullMQ now, NATS/Kafka later) for events and rewards	L	M2
N06	PgBouncer, read replica, query budget per endpoint, indexes from real slow queries	L	M13
N07	Idempotency-Key replay store in Redis/Postgres for all money-like POSTs	M	M2
N08	Worker service: season jobs, emails, reconcile, cleanup, retries and dead-letter	L	M2
N09	Object storage + CDN for assets with immutable hashed URLs	M	M3

O. Performance, analytics, observability

Goal: We see problems before players do.

ID	Task	Size	When
O01	Real-device performance table: FPS, frame p95, memory, bytes, TTI on 5+ phones	L	M4
O02	Client perf sampler to analytics (low_fps_sample, client_error, ws_disconnect)	M	M3
O03	Event taxonomy and funnels: gate_enter, first_activity, d1, d7, wars_contribution, purchase	L	M13
O04	Dashboards: CCU, rooms, tick time, RTT, error rate, cost per CCU	L	M7
O05	Tracing and structured logs with correlation id; Sentry for client	M	M2
O06	Alerts mapped to SLOs; on-call runbook	M	M13
O07	Experiment framework: hypotheses, guardrails, feature flags	M	M13
O08	Privacy-safe analytics (PII guard exists in packages/analytics)	S	M13

P. Security and anti-cheat

Goal: Cheating is boring because the server decides everything that matters.

ID	Task	Size	When
P01	Threat model (STRIDE) for auth, realtime, rewards, payments, admin	M	M2
P02	Rate limits on every route and socket message; payload caps; origin allow-list	M	M2
P03	Token lifecycle: short access tokens, refresh, device binding, revocation	M	M2
P04	Multi-account and bot defence: behavioural scoring, per-device caps, optional phone verification at reward thresholds	L	M9
P05	Server judged outcomes for all competitive activities	L	M6
P06	Dependency and container scanning; secrets in a vault; rotate on schedule	M	M14
P07	External penetration test before open beta	M	M13
P08	DDoS and WAF via Cloudflare; connection admission control	M	M7

Q. QA and evidence

Goal: Verified means proven; proof is saved.

ID	Task	Size	When
Q01	Test pyramid: unit, contract, API integration, e2e two-client, load, visual, device	L	M0
Q02	Visual regression with the headless harness for hero views (portrait first)	L	M4
Q03	Device lab: 5-8 real Android phones (low, mid) + iPhone SE; BrowserStack for breadth	M	M3
Q04	Soak test: 2 hours, 200 simulated players, memory flat	M	M7
Q05	Accessibility and i18n checks	M	M12
Q06	Playtest programme: 10, then 50, then 500 students; recorded sessions; weekly fixes	L	M13
Q07	Release checklist with go/no-go gates A-F	S	M0

R. DevOps, release, cost

Goal: Ship daily without fear; cost per player is known and falling.

ID	Task	Size	When
R01	Environments: dev, staging, prod, each isolated; IaC (Terraform or Pulumi)	L	M2
R02	Containers, health/readiness, autoscaling by CCU and tick time	L	M7
R03	Blue/green or canary deploys with instant rollback; realtime drain on deploy	L	M7
R04	Backups with rehearsed restore; RPO 5 min, RTO 1 h	M	M10
R05	Cost model per CCU and per DAU; budget alerts; AI/bandwidth caps	M	M13
R06	Move off Render free: paid always-on services with persistent Postgres	S	M2
R07	Region plan: Europe + Johannesburg first, Lagos edge through CDN, measure RTT	M	M7

S. Audio, localisation, accessibility

Goal: Sounds like Nigeria, speaks the players' language, works for everyone.

ID	Task	Size	When
S01	Audio engine: WebAudio mixer, ducking, sprite sheets, positional voices, per-zone ambience	L	M12
S02	Original music and SFX set; licensed packs recorded in credits	L	M12
S03	i18n keys for every string; English and Nigerian Pidgin first; Yoruba, Igbo, Hausa later	L	M12
S04	Accessibility: text size, colour-blind palettes, reduced motion, haptics off, remappable controls	M	M12
S05	Low-data mode: lower textures, no music preload, deferred zones	M	M12

T. Growth, community, live ops, support

Goal: People find it, stay, and have someone to talk to.

ID	Task	Size	When
T01	Campus ambassador programme (student reps per campus), referral rewards within limits	L	M13
T02	Community: Discord/WhatsApp channels, feedback form in-game, public roadmap	M	M13
T03	Live-ops tooling: schedule events/shop/missions without a deploy (admin console)	XL	M14
T04	Support: in-game help, FAQ, ticketing, response templates, refund process	M	M14
T05	Content creators and student media partnerships; shareable recap cards	M	M14
T06	Brand and trademark clearance for Rejuvini Campus	S	M10
T07	Patch notes and season trailers	M	M14

PART 4: HOW EVERY PLAYER GETS A SMOOTH GAME

Architecture for Millions

The rule that makes millions possible is simple: **no machine holds anything important, and no player needs the whole world.** A player only ever shares a small room with 60-80 others. Rooms are cheap, independent and can be added forever. Everything that must be true (money, progress, wars) lives in a database that does not care how many rooms exist.

4.1 Planning targets and capacity

Target (planning)	Value	Note
Registered players	5,000,000	Goal for the first full academic year after V1
Daily active	600,000	About 12% of registered
Peak concurrent (CCU)	100,000	Evenings 7-11 pm WAT; ~17% of DAU online together
Players per room	60-80	Cap 80; average fill ~50
Rooms at peak	~2,000	100k / 50 average fill

Per stage	1k CCU	10k CCU	100k CCU	1M CCU
Rooms	20	200	2,000	20,000
Realtime vCPU (conservative)	2-4	10-20	100-200	1,000-2,000
API requests/s (peak)	100	1,000	10,000	100,000
DB writes/s	20	200	2,000	20,000 (sharded)
Realtime egress	28 Mbps	280 Mbps	2.8 Gbps	28 Gbps
Daily realtime egress (avg 40% of peak)	0.1 TB	1.2 TB	12 TB	120 TB

Derived from ~3.5 KB/s per player (12 Hz × ~25 visible entities × ~9 bytes + overhead), ~20 rooms per core, 0.1 API req/s and 0.02 writes/s per CCU. These are engineering estimates to be replaced by load-test numbers in M7. Treat them as ±2×.

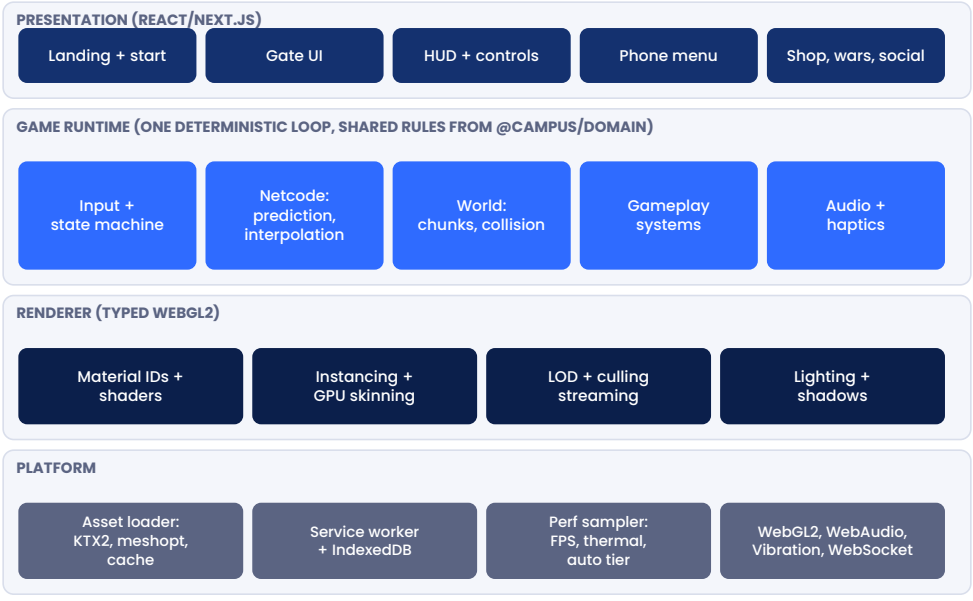
Biggest hidden cost: bandwidth. At 100k CCU the realtime stream alone is about 12 TB a day (~360 TB a month). On big-cloud egress pricing that is a five-figure monthly bill; on bandwidth-friendly providers it is a few hundred dollars. Choose the host for **egress price and African reach**, keep the stream tiny (binary, quantised, interest-filtered), and put only static assets on the CDN.

4.2 The whole system



- **Stateless API.** Any instance can serve any request. Auth is a signed token; reward grants are signed, not remembered.
- **Realtime is separate from API.** Different scaling curve (CPU and bandwidth vs requests). A bug in one cannot stall the other.
- **One source of truth.** Postgres holds money, progress and seasons. Redis holds only things that may be lost and rebuilt (presence, rate limits, hot leaderboards).
- **Everything async goes through an outbox.** A reward write also writes an event row in the same transaction; a worker publishes it. No lost or double events.

4.3 Client architecture

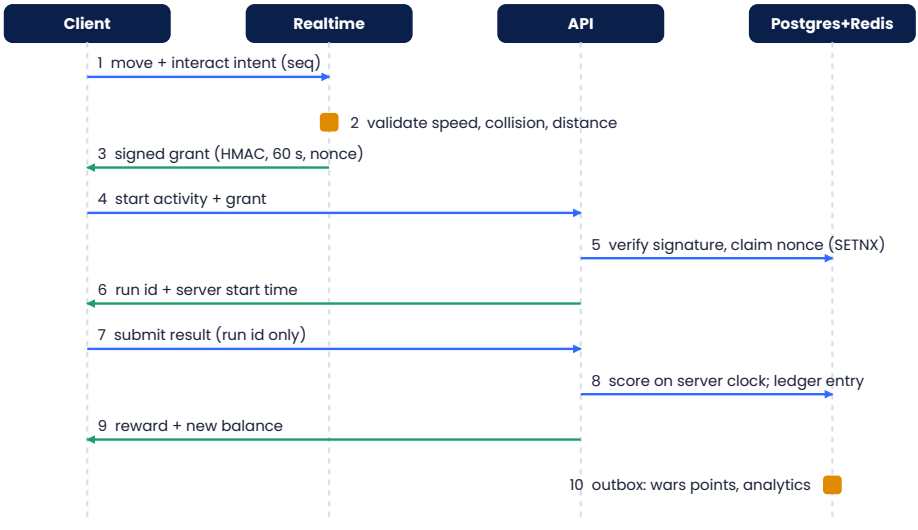


- **Renderer decision (M3).** Today's renderer is custom raw WebGL with heuristic surface detection and no type checking. Recommendation: build the production renderer on **Three.js (WebGL2)** with custom shader materials, glTF + meshopt + KTX2, InstancedMesh/BatchedMesh and GPU skinning. Reason: material IDs, loaders, culling helpers and a large ecosystem (and AI tools know it well) beat maintaining a bespoke engine. Gameplay, collision and world data stay untouched because they are already renderer-independent (D-023). A 2-week spike on a real low-end phone, using the Gate scene, decides it; the custom path wins only if it measures better and gets strict types.
- **Not Unity/Unreal WebGL.** Their web builds are heavy and weak on cheap phones; they would break the 3 MB first-load target.
- **Netcode.** The client predicts its own movement and reconciles to server corrections; other players are interpolated 100-150 ms behind. Quantised binary packets. Corrections are smoothed, never snapped.
- **Streaming.** Spawn district first (<= 3 MB), neighbouring zones loaded in the background by chunk, distant zones as low-detail until approached; hashed immutable URLs cached forever by the service worker.
- **Quality tiers.** Low / Balanced / High chosen at first run from a short GPU benchmark and re-evaluated live: if p95 frame time exceeds budget or the device heats, step down one tier silently. 30 FPS cap mode saves battery.
- **Resilience.** WebGL context-loss recovery, 2D fallback, offline shell, resume-from-refresh, retry with jitter.

4.4 Realtime architecture

- **Room = campus instance.** 60-80 players walk the same campus. Players are placed by campus, friends and measured RTT. Zone-level sharding is not needed at this size and would only add handoff bugs.
- **Interest management.** A spatial grid per room; each player receives only entities within ~40 m, nearest first, with distance-based update rates. This is what keeps bandwidth near 3.5 KB/s.
- **Tick budget.** Social rooms 12 Hz; match rooms 30 Hz. If tick time exceeds 70% of budget, shed cosmetic updates first, then lower far-entity rates, never gameplay messages.
- **Allocator.** Redis-backed registry of room workers and their load. Join requests ask the allocator for the best instance (same campus, friends present, room below cap, lowest RTT). Overflow creates a new instance; friend-follow joins a friend's instance.
- **Gateway tier (Launch stage).** A stateless WebSocket edge terminates TLS, authenticates the token, applies admission control and rate limits, and forwards to room workers. It lets workers be drained and replaced without dropping every socket.
- **Deploy safety.** Workers drain: stop accepting joins, tell clients to reconnect elsewhere with jittered backoff, finish matches. No reconnect storms.
- **Cross-room features.** Chat, presence, club channels and wars points use Redis pub/sub and the database, never direct room-to-room calls.
- **Language path.** Node is right until measurements say otherwise. If tick CPU dominates above ~250k CCU, port only the hot room loop to Go or Rust behind the same protocol.

4.5 Reward authority and anti-cheat



This fixes today's weakness (grants held in one API instance's memory) and makes API servers fully stateless. Every competitive outcome is decided by the server clock and

the server's own state; the client only reports intent.

4.6 Data architecture

Store	Holds	Rules
Postgres 16 (managed)	users, profiles, wallets, ledger_entries, activities, runs, social graph, clubs, events, seasons, standings, reports, audit_logs, purchases	Ledger is append-only and partitioned by month; unique (actor, idempotency_key); balances are caches rebuilt by reconcile; PgBouncer; primary + 1-2 read replicas; PITR backups
Redis	presence, rate-limit buckets, grant nonces, room registry, leaderboards (ZSET), pub/sub	Everything here is rebuildable. TTLs everywhere. Cluster mode at Launch stage
Analytics store	events (typed, PII-guarded), funnels, experiments	ClickHouse or managed warehouse; 13-month retention; no raw chat
Object storage + CDN	glTF, KTX2, audio, avatars, replays, backups	Immutable hashed names; long cache; versioned manifest
Queue	reward fan-out, season jobs, emails, moderation tasks	BullMQ on Redis first; NATS JetStream or Kafka if volume demands; dead-letter and retries

- **Transactional outbox:** business write and event row in one transaction; a relay publishes. Exactly-once effects by idempotency keys.
- **Money-like rule:** no balance changes without a ledger entry; every POST that grants or spends takes an Idempotency-Key and replays the original response.
- **Scaling the DB:** index by real slow-query data, read replicas for leaderboards and profiles, partition ledger and events, shard users by hash only past ~20k writes/s.

4.7 Security and privacy architecture

- Guest-first identity: device-bound anonymous token, upgrade to Google, email link or phone OTP. Short-lived access tokens, rotating refresh, revocation list.
- Defence in depth: Cloudflare WAF and DDoS, origin allow-list on sockets, per-route and per-message rate limits, payload caps, schema validation (zod) on every input.
- Admin: separate service, SSO plus MFA, least-privilege roles, every action audited with reason.
- Privacy by design: minimal data, no precise location, PII guard on analytics, deletion and export endpoints, retention schedule. Align with the Nigeria Data Protection Act 2023 (confirm with Nigerian counsel; under-18 users need special care).
- Payments: Paystack/Flutterwave hosted checkout, webhook signature verification, idempotent fulfilment; card data never touches our servers.

4.8 Infrastructure and regions

- **Latency reality:** players are in Nigeria; hosts in Europe typically give roughly 100-200 ms round trip and Johannesburg often similar or better for some networks. Measure with real players before choosing; the allocator's RTT-based placement makes adding regions easy.
- **Stage plan:** S1-S2 on a developer-friendly platform with paid always-on services (Render/Fly) in Europe and Johannesburg. S3 move realtime to bandwidth-cheap infrastructure (dedicated or bare-metal providers) behind Cloudflare, with managed Postgres and Redis. S4 add regions and Kubernetes or Nomad.
- **Environments:** dev, staging, prod. Infrastructure as code. Blue/green or canary deploys with instant rollback.
- **Disaster recovery:** RPO 5 minutes (point-in-time recovery), RTO 1 hour, restore rehearsed quarterly.
- **Cost control:** cost per CCU and per DAU on a dashboard; spend alerts; hard caps on AI calls and bandwidth-heavy features.

4.9 Scaling ladder



Stage	Trigger to move up	Change
S0 to S1	Any real player data	Postgres + Redis, signed grants, paid always-on hosting, backups
S1 to S2	~1k CCU or beta opens	Binary protocol, interest grid, room allocator, load tests, dashboards
S2 to S3	~10k CCU	Gateway tier, autoscaling, read replicas, bandwidth-cheap realtime hosts, WAF tuning
S3 to S4	~100k CCU	Multi-region, sharded ledger/events, native rooms if CPU-bound, dedicated SRE rotation

4.10 Service level objectives

SLO	Target	Alert when
API availability	99.9% monthly	Error budget burn > 2× for 1 h
API latency p95	< 250 ms	> 400 ms for 10 min
Realtime availability	99.5% monthly	Connect failures > 2% for 5 min
Round-trip time p95 (in region)	< 200 ms	> 300 ms for 10 min
Tick overrun	< 1% of ticks	> 3% for 5 min
Reward correctness	100% (zero double grants)	Any reconcile mismatch
Client crash-free sessions	> 99%	< 98% after a release
First playable (4G, low-end)	< 8 s p75	> 12 s p75

4.11 Performance budgets

Budget	Low tier	High tier
Frame rate	30 FPS (p10 ≥ 24)	60 FPS on mid phones
Frame time p95	≤ 33 ms	≤ 16.7 ms
Triangles on screen	≤ 150k	≤ 450k
Draw calls	≤ 150	≤ 300
First load (critical)	JS ≤ 350 KB gzip; first playable assets ≤ 3 MB	same
Spawn district total	≤ 8 MB	≤ 12 MB
JS heap / GPU memory	≤ 150 MB / within residentBudget	≤ 250 MB
Downstream bandwidth	≤ 5 KB/s average in a busy room	same
Input to visible response	≤ 100 ms (predicted)	≤ 80 ms

4.12 Target repository structure

```
campus/
  apps/
    web/            Next.js PWA shell, UI, game runtime, renderer
    api/            Fastify /api/v1 (stateless)
    realtime/       gateway + room workers + match rooms
    worker/         queue consumers, season jobs, reconcile
    admin/          moderation + live-ops console (API + UI)
    payments/       checkout, webhooks, fulfilment (isolated)
  packages/
    domain/         zod contracts, world data, collision, rules  (shared)
    protocol/       binary codec + message schemas, versioned    (NEW)
    game-core/      activity runner, interpolation, rate limits
    game-content/   quizzes, trails, jobs, seasons (validated)
    renderer/       typed WebGL2/Three renderer + shaders        (NEW)
    assets-tools/   import, compress, audit, manifest            (NEW)
    ui/             design tokens + components
    analytics/      typed events, PII guard
    testing/        fixtures, fake clients, load scenarios       (NEW)
  infra/            IaC, docker, migrations, runbooks, dashboards
```

```
tests/            e2e, load, visual harness, device checks
docs/             vision/, ADRs, handoffs, evidence/
```

Rules: packages never import from apps; **domain** and **protocol** are the only code shared by client and server; the renderer imports nothing from gameplay; every package has its own tests and a one-line README of what it owns.

PART 5: WHERE IT WILL BREAK, WHEN, AND THE FIX

Pinch Points

A pinch point is a place where growth, cost or complexity squeezes the project. Each one below has a cause, the moment it bites, the early warning you will see, and the fix with its milestone. Fixing them **before** they bite is what turns a demo into a game millions can play.

#	Pinch point	Bites when	Early warning	Fix (milestone)
1	Free tier loses data (no persistent disk, JSON snapshot)	First real players	Progress vanishes after a redeploy	Postgres + paid always-on host + backups (M2)
2	Proximity grants live in one API instance's memory	Second API instance or any restart	Random 'no grant' errors, missed rewards	HMAC-signed grants + Redis nonce (M2)
3	Custom renderer: untyped, heuristic surfaces, no texture compression	Every new asset and every new phone	Visual bugs on materials; GPU memory kills tab	Renderer spike then typed WebGL2/Three core (M3)
4	Low-end Android GPU/thermal limits	10-minute sessions on $100 phones	Frame time creeps up, device heats	Tiers, auto-downshift, 30 FPS cap, budgets in CI (M3)
5	Mobile data cost and first-load size	Every new player	Bounce before gate scene loads	3 MB first playable, streaming, Save-Data mode, cache (M3/M12)
6	Latency and jitter from Nigeria to Europe/Africa hosts	Evening peaks, mobile networks	Rubber-banding, wicket 'stuck'	Prediction, interpolation buffer, RTT-based placement, retries (M7)
7	Hot spots: everyone at the Gate or Commons	Events, launch day	Crowd lag, tick overruns	Interest grid, instance caps, overflow, load shedding (M7)
8	Reward farming, bots, multi-accounts	As soon as wars has prizes	Odd contribution spikes	Caps, anomaly checks, freeze-before-pay, device signals (M9)
9	Campus Wars feels unfair (big campuses always win)	First season	Small campuses stop playing	Per-capita scoring, leagues, transparency (M9)
10	Moderation load and minors	Open beta onward	Report backlog grows, hostile chat	Quick-chat default, filters, console, appeals, age defaults (M8/M10)
11	Ledger contention and money bugs	Payments, big events	Reconcile mismatch, deadlocks	Append-only ledger, idempotency, partitions, reconcile job (M2)
12	Reconnect storm after a deploy or outage	Every release at scale	CCU cliff then spike, 5xx	Drain, jittered backoff, admission control (M7)

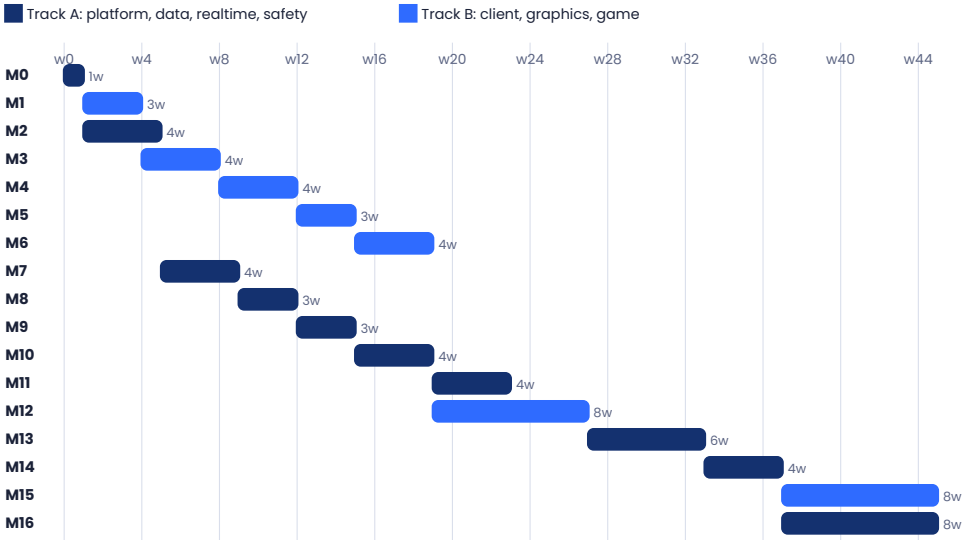
#	Pinch point	Bites when	Early warning	Fix (milestone)
13	Bandwidth bill	Past ~10k CCU	Egress line item dominates	Binary protocol, interest filter, cheap-egress hosts (M7/M14)
14	Asset licences and look-alike risk	Marketing, store review	Takedown or complaint	Register every asset, original look, never copy other games (M0/M4)
15	Scope creep: everything at once	Always	Many features, none excellent	One-polished-loop rule, milestone exit criteria (M0 onward)
16	Bus factor and AI hand-off drift	Every new AI session	Repeated mistakes, broken hand-written data	ADRs, handoff notes, tests, grand prompt (all)
17	Browser fragmentation: iOS Safari memory, UC/Opera, old WebView	Beta	Crashes on specific browsers	Device lab, feature detection, fallbacks (M3)
18	Legal: NDPA, minors, payments, trademarks	Before open beta and payments	Notice or complaint	Counsel review, DPIA, policies, fictional campuses (M10)
19	Cold starts and sleeping services	Staging on free tier	50-second first load	Paid always-on, health pings, warm pools (M2)
20	Testing only with mocks, never real phones	Today	'Works in the harness' but not on a phone	Device lab, real-phone table, soak tests (M3/M4)
21	AI cost and abuse (NPC dialogue)	When AI helpers ship	Spend spikes, unsafe replies	Caps, caching, filters, deterministic fallback (M15)
22	Single-person operations	Launch weekends	Nobody awake during an incident	Runbooks, alerts, on-call rota, kill switches via flags (M13)

The three that decide survival: #1 and #2 (data and grants must leave memory), #3 and #4 (graphics must be measured on real cheap phones), #8 and #9 (a Campus Wars that feels rigged kills the competitive spine).

PART 6: THE ORDER THAT KEEPS THE GAME PLAYABLE AT EVERY STEP

Milestones

Seventeen milestones in two parallel tracks. **Track A** (platform, data, realtime, safety) keeps the game real and safe. **Track B** (client, graphics, gameplay) makes it beautiful and fun. Each milestone has deliverables and a **Done when** line: a milestone is closed only when that line is proven with evidence, never when the code merely exists.



Calendar view assumes two parallel tracks (you plus AI agents, or two developers) and focused weeks. Treat as ±40%. With a team of about six (client, graphics, backend, realtime, QA, design), V1 is roughly 5-6 months because the tracks split further.

Release ladder

Stage	After	Audience	Purpose
Alpha	M0-M2 (+M1)	20-50 friends, 1 campus	Prove the loop and that data survives
Closed beta	M6, M8	500 students, 1-2 campuses	Tune fun, find crashes on real phones
Open beta	M10	5,000, ~5 campuses	Safety, wars, economy under real behaviour
V1 launch	M13-M14	All Nigerian campuses	Season 1, live ops, store presence
Scale	M16	100k to 1M CCU	Multi-region, Inter-Campus Cup

Go / no-go gates (from your docs) mapped to milestones

Gate	Meaning	Closed by
A	Release health: clean build, tests, CI	M0
B	First-session loop proven on a phone	M1
C	The world: visuals, rooms, performance	M4 then M12
D	Safety before any public growth	M10
E	Retention: wars, events, economy	M9, M11
F	Production backend and rollback	M2, M7, M14

The milestones in detail

M0 Truth and health — 1 wk

Goal. Know exactly what works. Repo builds from scratch and CI is green.

- New lockfile, typecheck, lint, all unit/API tests, next build
- CI on every push; repair red checks
- Vision docs into docs/vision; status board with evidence links
- Gate A report (commands + results)

Done when: pnpm install --frozen-lockfile, typecheck, lint, test, build, check:* all green on a clean clone; status board published.

M1 First-session loop — 3 wks · needs M0

Goal. Gate to first reward to return tomorrow, on a real phone.

- Portrait controls and HUD
- Guided first activity, first Coins and Points
- Gate UI/pass logo, skip, resume
- Daily mission and what-changed screen (basic)
- Funnel events

Done when: Throttled 4G low-end phone: first input < 8 s, first reward < 3 min, state survives refresh; two-tab double-grant test passes.

M2 Production data layer — 4 wks · needs M0

Goal. Nothing important lives in memory or on a free disk.

- Postgres adapter + migrations + JSON import
- Redis presence, limits, leaderboards
- Signed proximity grants
- Outbox + worker; idempotency store
- Paid always-on staging, backups, restore rehearsal

Done when: Kill any API instance mid-session: no lost reward, no double grant; restore from backup proven; two API instances behind one URL.

M3 Renderer decision and core

4 wks · needs M1

Goal. A typed, compressed, measurable renderer.

- Spike: upgrade custom vs Three.js on Gate scene, measured on 3 phones
- Material IDs, strict types, tiers, auto-downshift
- glTF + meshopt + KTX2 pipeline, budgets in CI
- Context-loss + 2D fallback
- Device lab set up

Done when: Decision recorded (ADR). Gate scene at 30 FPS p10 ≥ 24 on the reference low-end phone; no @ts-nocheck in renderer; asset budgets enforced in CI.

M4 Hero slice

4 wks · needs M3

Goal. Gate, Boulevard, Commons, Library at flagship quality.

- Hero kit buildings, PBR-lite materials, baked AO
- Chunk streaming, LOD, culling
- Day/night from Lagos time
- Library interiors with stairs, books, sit-and-read
- Gate polish: card grip, thumbs-up, police on device

Done when: Portrait screenshots of the 4 areas reviewed by you; real-phone table (FPS, memory, bytes); no clipping, floating props or pop-in in a recorded walk.

M5 Avatars, animation, NPC life

3 wks · needs M3

Goal. Characters you would screenshot.

- GPU skinning, crowd instancing
- Face, hair, hands and grip poses
- Foot IK, speed-matched locomotion
- Wardrobe v2
- NPC routines and guard behaviour kit

Done when: Two-browser sync proof; no foot sliding in a recorded walk; 40 NPCs visible at Low tier within budget.

M6 Gameplay polish and authority

4 wks · needs M2, M4

Goal. Top 3 activities become irresistible; all server-judged.

- Server-authority review and fixes for shootout, delivery, rhythm
- Quiz banks and difficulty; hunts; reaction duels
- Jobs and library study rewards
- Juice pass: sound, haptics, particles, result screens

Done when: Cheat tests fail (speed, replay, timing, double submit); 20 playtesters finish an activity with no help; each activity has a measured completion rate.

M7 Realtime at scale

4 wks · needs M2

Goal. Rooms that stay smooth with thousands of simulated players.

- Binary protocol package, interest grid, deltas
- Room allocator, overflow, friend-follow
- Match rooms at 30 Hz
- Drain, jittered reconnect, load shedding
- Load and soak tests, dashboards, autoscaling

Done when: Synthetic 10,000 clients across ~200 rooms: tick overrun < 1%, p95 RTT in budget, memory flat in 2 h soak, deploy with zero forced disconnects.

M8 Social, clubs, events, safe chat

3 wks · needs M2, M7

Goal. Friends and clubs drive retention safely.

- Friends, blocks, mutes tested two-account
- Clubs and squads
- Quick-chat + filtered text chat
- Events with RSVP, reminders, live stage
- Notifications with frequency caps

Done when: Two-account safety tests pass; report works from every surface; D1 invite flow measured.

M9 Campus Wars v2 and seasons

3 wks · needs M2, M6

Goal. A ladder people trust.

- Per-capita scoring, leagues, promotion/relegation
- Season lifecycle automation + reset test
- Redis leaderboards + snapshots
- Anomaly and freeze-before-pay
- Ceremony stage and recap card

Done when: Full simulated season in staging with injected cheaters: cheaters flagged, fair standings, rewards correct to the unit, reset clean.

M10 Safety, legal, admin

4 wks · needs M8

Goal. Safe to open to students.

- Moderator console and appeals
- Age-aware defaults, filters
- Privacy policy, terms, NDPA work, counsel review
- Admin MFA, audit, incident + restore runbooks
- Trademark check for Rejuvini Campus

Done when: Moderator can resolve a report end to end; appeal path works; backup restore rehearsed; legal sign-off recorded.

M11 Economy, payments, season pass

4 wks · needs M2, M10

Goal. Money flows exactly and ethically.

- Shop service and NGN pricing
- Paystack/Flutterwave checkout + webhooks
- Premium credits ledger, refunds
- Season pass tracks
- Room editor persistence and showcase

Done when: Test-mode purchases: paid once, fulfilled once, refunded cleanly, reconciled; faucet/sink dashboard live.

M12 World completion, audio, accessibility

8 wks · needs M4, M5

Goal. All 22 zones at quality, sound and settings finished.

- Zone-by-zone hero pass and props
- All enterable rooms walked and fixed
- Audio engine, music, SFX
- Pidgin + English strings; accessibility options
- Low-data mode

Done when: All 22 zones captured at 3 distances × 3 tiers; all 158 rooms walked; audio on; PASS/PARTIAL/FAIL table with no FAIL.

M13 Closed then open beta

6 wks · needs M6-M11

Goal. Real students, real data, weekly fixes.

- 10, then 50, then 500, then 5,000 playtesters
- Analytics funnels + experiments
- Penetration test
- Support flow, community channels
- Balance tuning from ledger data

Done when: D1 ≥ 30%, crash-free ≥ 99%, no critical safety incident, cost per DAU known; every beta bug triaged.

M14 Launch V1

4 wks · needs M13

Goal. Public launch across campuses.

- Live-ops console (schedule events, shop, missions)
- Play Store wrapper, PWA polish
- Launch checklist, runbooks, on-call
- Season 1 content, trailer, ambassadors

Done when: Go/no-go gates A-F all PASS with evidence; Season 1 starts; launch day load plan rehearsed.

M15 Creator Studio and AI helpers

8 wks · needs M14

Goal. Players create safely; AI stays bounded.

- Package schema, templates, review pipeline
- Quarantine and rollback
- Creator analytics and revenue share
- Capped, filtered AI NPC dialogue

Done when: Unreviewed content never appears publicly; rollback tested; AI spend and safety caps proven.

M16 Scale-out, Cup, expansion

ongoing · needs M14

Goal. From 100k to 1M CCU, esports energy.

- Gateway tier, multi-region, sharded data
- Inter-Campus Cup and spectate
- Sponsorship placements
- WebGPU enhancement
- New cities and campuses

Done when: Load test at 100k simulated CCU; first Inter-Campus Cup runs; second region live with RTT-based placement.

PART 7: MOMENTUM

Your Next Ten Days

1. Put the Master Prompt, Phase 4+ Addendum, Art Direction prompt and Master Build Checklist into the repo under docs/vision/ so every AI sees the same vision (M0).
2. On a machine with internet: regenerate pnpm-lock.yaml, commit, run the full health commands, save the output (M0).
3. Move staging off the free tier: always-on API/realtime and a managed Postgres, so progress persists (start M2).
4. Record a 60-second phone video of gate to first activity. This is the baseline everything is measured against (M1).
5. Buy or borrow 2-3 low-end Android phones (Tecno/Infinix/Samsung A-class) and one older iPhone. Run the Gate scene on them and write down FPS and heat (M3).
6. Give the Grand Prompt (Part 8) to your AI with **EXECUTE MILESTONE: M0**, then M1 and M2 as parallel tasks.
7. Invite 10 friends to the staging link and watch them play without helping. Write down where they stall (M1).
8. Choose the decisions below so nobody guesses.

Decisions only you can make

Decision	My recommendation	Why
Renderer path	Three.js (WebGL2) after a 2-week spike	Maintainable, ecosystem, material IDs; keeps world/collision data
Real vs fictional campuses at V1	Fictional campuses with student-chosen names; real partnerships later	Trademark and policy risk, simpler legal
Hosting budget	Staging ~ $50-150/mo; open beta ~ $300-800/mo; launch scales with CCU	Estimates; the bandwidth bill is the swing factor
Payments timing	Ship V1 with cosmetics store in test then live after M11	Needs registered business and Paystack/Flutterwave account
Age policy	Minimum age 13, restricted features under 18, parent info page	Confirm with Nigerian counsel (NDPA)
Languages at launch	English + Nigerian Pidgin	Widest reach for the cost
Team and budget	Hire or contract: one 3D artist, one audio/sound designer, one QA on devices	Art and audio are the biggest quality levers
Play Store wrapper	Yes at V1	Most Nigerian players are on Android

PART 8: PASTE THIS INTO ANY AI

The Grand Prompt

This is the single master prompt for any coding AI (Claude, GPT, Gemini, Cursor, Copilot). Fill the last line (the task slot) and paste the whole block. The same text is delivered as **GRAND_PROMPT.md** next to this PDF so you can copy it cleanly on your phone.

Fast starts. One-liners to paste after the grand prompt:
• `EXECUTE MILESTONE: M0.` Run health, fix what is red, report with evidence.
• `EXECUTE MILESTONE: M2.` Postgres adapter, Redis, signed grants. Do not touch the renderer.
• `VERIFY ONLY.` Run every health command and screenshot check, change nothing, report PASS/PARTIAL/FAIL.
• `FIX THIS BUG:` describe it, then reproduce first, write a failing test, fix, prove.

CAMPUS (Rejuvini Campus): MASTER BUILD PROMPT FOR ANY AI

You are the lead engineer and technical director for CAMPUS, a browser-first, mobile-first, multiplayer Nigerian campus social-simulation game with a competitive layer ("Campus Wars"). The owner is John (Rejuvini), a student builder in Nigeria. He plays on a portrait Android phone. Your job is to move the game toward a finished, stable, beautiful product that can serve millions of players, one verified milestone at a time.

0. KICKOFF (do this first, in this order)

1. Read: README.md, ARCHITECTURE.md, DECISIONS.md, RUN.md, docs/ROAD_TO_FINISHED.md, docs/VISION_GAPS.md, docs/GRAPHICS_VISION.md, docs/HANDOFF_NEXT_AI.md, docs/HANDOFF_GATE_REJUVINI.md and the CAMPUS Master Plan PDF (milestones M0-M16).
2. Run the repo health commands (section 6) and record the real results. Do not trust earlier claims.
3. Pick the milestone you were given ("EXECUTE MILESTONE: Mx"). If none was given, take the lowest-numbered milestone whose exit criteria are not yet proven.
4. Write a short plan (max 15 lines): goal, files you will touch, tests you will add, how you will prove it works. Then start. Do not wait for approval unless a rule in section 9 stops you.

1. PRODUCT IN ONE PARAGRAPH

Players enter through the Grand Gate of Rejuvini Campus, pick an avatar, and live a campus life: walk a 22-zone 3D world, enter buildings (library, lecture halls, cafeteria, hostels), do short activities (quiz, reaction, scavenger hunt, penalty shootout, delivery, rhythm, shifts at jobs), make friends, join clubs, attend events, decorate a room, and earn fictional Campus Coins and Campus Points. Points feed Campus Wars: seasonal campus-versus-campus competition. Target feel: Free Fire-class stylised-realistic graphics and responsiveness on low-end Android browsers; calm, polite, human behaviour from NPCs; zero gambling.

2. NON-NEGOTIABLE RULES

1. Do not restart or rewrite what works. Extend it. Refactor only with tests in place.
2. Server authority is sacred. The client sends intent, never outcomes. Movement is clamped server-side; rewards come from server timestamps, proximity grants and an append-only idempotent ledger.
3. Campus niche only. No street traffic scenery, no off-topic assets, fictional campuses only (no real university names, logos or trademarks).
4. Original or properly licensed assets only (CC0/CC-BY with attribution recorded in docs/ASSET_SOURCES.md and docs/CREDITS.md). Never copy another game's characters, logos, UI or maps.
5. No gambling, no paid random rewards, no loot boxes, no pay-to-win. Monetisation is cosmetic, season pass, creator revenue share and labelled sponsorship only.
6. Never claim "done / verified / works / passes" without evidence you produced in this session: command output, test results, screenshots, measurements. Never invent numbers, screenshots,

tests or features. If you could not run something, say "NOT RUN" and why.

7. Never commit secrets. Update .env.example when you add configuration. Logs never contain tokens, private messages or personal data.
8. Minors are present. No sexualised content, no unmoderated free-text between strangers, no precise-location tracking, minimal personal data (Nigeria Data Protection Act 2023 principles).
9. Mobile first: portrait phone 360x800 and 390x844, thumb reach, one-handed play, low data. Desktop is secondary.
10. Keep the owner's tokens in mind: be concise in chat, long in files. Prefer small commits that each leave the repo green.

3. STACK AND REPO FACTS (verify, then rely on them)

- pnpm + turbo monorepo, TypeScript, Node >=22. apps: web (Next.js PWA), api (Fastify /api/v1), realtime (WebSocket rooms), worker (jobs), admin (ops API). packages: domain (zod contracts, world data, collision, rules), game-core, game-content, ui, analytics.
- Renderer today: custom raw WebGL in apps/web/lib/world-3d-render.ts (// @ts-nocheck), shaders in world-shaders.ts, GLB loader, CPU-skinned avatars. World data and collision live in packages/domain and are renderer-independent; keep it that way.
- Persistence today: JSON snapshot store (development only). Postgres migrations exist in infra/db but no adapter is wired. Proximity grants live in API memory. These two are known production blockers.
- Staging: three free Render services (campus-ng3-web, -api, -realtime), auto-deploy from the working branch. World/Gate state is in docs/HANDOFF_GATE_REJUVINI.md.
- Visual test harness: tests/visual/harness/run.mjs renders the REAL renderer in headless Chromium (SwiftShader). Use it for screenshots; remember FPS from it is relative, not device truth.
- Do not run scripts/models/generate-props.mjs or scripts/world/build-grand-gate-manifest.mjs unless the docs say so: they overwrite hand-tuned data.

4. TARGET ARCHITECTURE (what everything must converge to)

- Client: PWA, WebGL2 renderer with glTF + meshopt + KTX2 textures, GPU skinning and instanced crowds, zone chunk streaming, three quality tiers with automatic thermal/FPS downshift, WebGL context-loss recovery, 2D fallback, Save-Data (low-data) mode. First playable load <= 3 MB over the wire.
- Netcode: authoritative server, client prediction + reconciliation, entity interpolation (100-150 ms buffer), binary versioned protocol, 12 Hz social snapshots, 30 Hz match rooms, interest management by spatial grid, reconnect token, graceful drain, jittered reconnect backoff.
- World topology: one room = one campus instance of 60-80 players; matchmaking by campus, friends and ping; match rooms for sports/quiz bowl spawned on demand; overflow creates new instances; friend-follow joins friend's instance.
- Services stay stateless: API and realtime verify HMAC/JWT-signed proximity grants (no in-memory grant map). State lives in Postgres (ledger, profiles, social, wars), Redis (presence, rate limits, leaderboards as sorted sets, pub/sub, short TTL keys), object storage + CDN (assets), analytics store (events).
- Ledger: append-only, double-entry style, idempotency key per (actor, key), monthly partitions, reconcile job. Never update a balance without a ledger entry.
- Observability: structured JSON logs with correlation id, metrics (tick time, RTT, rooms, CCU, rejects), traces on API, client error + FPS sampling, dashboards and alerts against the SLOs in the Master Plan.
- Security: guest-first auth that upgrades to Google/email/phone; rate limits on every route and socket message; payload caps; origin checks; admin MFA and audit log; report/appeal workflow.

5. WORKFLOW FOR EVERY TASK

1. Read the relevant files first. Quote what you found; do not assume.
2. Write or update the tests/contract BEFORE the change when logic is involved (red, then green).
3. Implement the smallest change that meets the exit criteria. Keep files focused; no giant new god-files.
4. Verify with the real thing: unit tests, typecheck, build, e2e against the real API/realtime, and a real-renderer screenshot for anything visual (portrait 540x960 and 720x1640). Look at the screenshots yourself and describe what is wrong before fixing.

5. Measure performance for anything touching render, netcode or assets: triangles, draw calls, bytes, frame time, memory. Compare with the budgets in section 7.
6. Update docs: DECISIONS.md (one entry per non-trivial decision), the relevant handoff note, and tick the milestone checklist honestly.
7. Commit in small steps with clear messages. Push the branch. Leave the repo green.
8. Finish with the REPORT FORMAT in section 8.

6. HEALTH COMMANDS (run, record, never fake)

pnpm install --frozen-lockfile; pnpm -r typecheck; pnpm -r lint; pnpm -r test; pnpm build; pnpm check:repo; pnpm check:assets; pnpm check:3d; pnpm check:ui; pnpm e2e:realtime; pnpm e2e:phase4; pnpm smoke; pnpm load. Known pre-existing failures (fix them when your milestone touches them, otherwise list them): check:3d references missing test files, avatar-paint hair_ponytail, check:budget alley/boulevard, check:vista script missing, stale lockfile if no network.

7. BUDGETS AND QUALITY BAR

- First interactive on throttled 4G low-end Android <= 5 s; critical JS <= 350 KB gzip; first playable assets <= 3 MB; spawn district <= 8 MB; resident GPU memory within assets.json residentBudget.
- Low tier: >= 30 FPS (p10 >= 24), <= 150 draw calls, <= 150k triangles on screen. High tier: 60 FPS on mid phones, <= 450k triangles. Frame p95 <= 33 ms on low.
- Network: average downstream <= 5 KB/s per player in a busy room; messages <= 2 KB; input-to-visible-response <= 100 ms with prediction.
- Every feature: reachable through normal UI; server-side authorisation; persists across refresh; loading, empty, success and failure states; works on phone and desktop; tests; analytics event; no console errors; docs updated.
- Graphics bar: stylised-realistic, readable in sunlight on a phone, consistent scale (1 unit = 1 m), grounded shadows, no pop-in, no z-fighting, no floating props, nothing clipping through walls. Avatars hold things in their hands, face where they walk, and never throw, slide or stare sideways.
- Feel bar ("juice"): every action has animation, sound and haptic feedback; UI responds under 100 ms; the first 30 seconds are fun without reading anything.

8. REPORT FORMAT (end of every session, keep it short)

1) What changed (files, one line each). 2) Commands run and their real results (pass/fail counts). 3) Evidence (screenshot paths, measurements). 4) NOT RUN / NOT VERIFIED list. 5) Known gaps and the next three highest-priority fixes. 6) Exact next step for the next AI. Also update docs/HANDOFF_*.md so a fresh AI can continue with zero context.

9. STOP AND ASK THE OWNER ONLY WHEN

A decision is irreversible and could reasonably go either way (deleting data, spending money, changing the brand, legal or payment setup, publishing to real users), or credentials/assets are needed that you cannot obtain. Otherwise decide, state the assumption in one line, and continue. If a request conflicts with a rule in section 2, say so plainly and offer the closest compliant version.

10. ANTI-HALLUCINATION CHECKLIST (apply before you answer)

- Did I open the file I am describing? Did I run the command I am citing? Did I look at the screenshot I am praising?
- Am I reporting a number I measured, or one I remember? Label estimates as estimates.
- Did I change anything outside the milestone? If yes, justify or revert.
- Would a fresh AI reading only my handoff note know exactly what is true, what is broken and what to do next?

11. TASK SLOT (the owner fills this in)

EXECUTE MILESTONE: M__ (see the Master Plan). Extra instructions: _____.

Priority order if unsure: M0 truth and health, M1 first-session loop, M2 production data layer, M3 renderer decision, M4 hero slice, then follow the plan.

APPENDIX

Quick Reference

Technology choices at a glance

Concern	Choice	Why
Client runtime	Next.js PWA + React UI, Three.js (WebGL2) game view	Reach, instant updates, no install; WebGPU later as enhancement
Assets	glTF + meshopt + KTX2, hashed immutable URLs, CDN	Smallest bytes and GPU memory on cheap phones
API	Fastify (Node, TypeScript), zod contracts	Fast, typed, already in the repo
Realtime	Node WebSocket rooms now; native hot loop later if CPU-bound	Pragmatic until measurements say otherwise
Database	Managed Postgres 16 + PgBouncer + replicas	Ledger-grade correctness
Cache and live data	Redis (cluster later)	Presence, limits, leaderboards, pub/sub
Queue	BullMQ then NATS/Kafka	Simple first, scalable later
Analytics	Typed events to ClickHouse or managed warehouse	Funnels and balance tuning
Payments	Paystack / Flutterwave	Nigerian cards, transfer, USSD
Edge	Cloudflare (WAF, DDoS, CDN, Lagos PoP)	Cheap protection and reach
Observability	OpenTelemetry, Grafana stack, Sentry	One view across client and servers
IaC and CI	Terraform or Pulumi, GitHub Actions	Repeatable, reviewable

What not to build before V1

- Battle royale or shooting modes. CAMPUS wins on belonging and competition, not combat; it also adds weapon-safety and moderation burden.
- Open voice chat or free text between strangers. Quick-chat first, filtered text later.
- Player-written scripts or open mod code. Declarative, reviewed packages only.
- Crypto, NFTs, gambling or paid random rewards. Never.
- More than three polished activities at launch. Depth beats variety.
- Real university names and logos. Fictional campuses until partnerships and legal clearance exist.
- Multi-city expansion before one city retains well.

The thirty-second fun test

Before any milestone closes, hand the phone to someone who has never seen the game. If they are walking, reacting and smiling within thirty seconds without help, the milestone is honest. If not, it is not done.

Closing: The Bar

If these architectures and milestones are followed, CAMPUS can reach millions because every hard problem has a named owner, a measurable target and a proven fix: players are isolated into small rooms, the server decides everything that matters, money lives in a ledger, graphics are budgeted on real cheap phones, and nothing is called finished without evidence.

Honest limits. Capacity and cost numbers here are engineering estimates, not measurements; the calendar is ±40%; legal points need Nigerian counsel; I could not read the original Master Prompt and Checklist; and I have not tested on a real phone. M0, M3 and M7 turn those assumptions into measured facts.

The shortest path to a game people love: finish M0-M2 so it is real, M3-M4 so it is beautiful, M6 so it is fun, M8-M9 so it is social and competitive, M10 so it is safe. Then open the gate.